

Paritran technical white paper

A deterministic investigation core with a fenced language layer

From complaint to conviction

TECHNICAL WHITE PAPER

Category 2 student team | ID 10_338

PS-69EEFE4F8CD1C | Cyber Crime Branch, Ahmedabad City

Truth boundary

Working prototype measured on synthetic data. No real deployment, external institution validation, autonomous legal decision, or production admissibility claim is made.

Canonical claim source SHA-256

800091205ad1e7c2a22a7add37f8d8fcfa1e5094f8737806bea56f5229fbc357

Abstract

Paritran is an on-premise cyber-fraud investigation prototype that links synthetic complaints into candidate mule networks, reconstructs a directed money trail, maps cited legal sections through agreement and abstention, gates drafted claims against source text, and records actions in a tamper-evident custody chain. The system is designed to assist an officer, never to decide guilt or certify admissibility.

The canonical synthetic run contains 297 complaints and six planted networks. The measured linkage precision, recall, and F1 are 0.957, 0.966, and 0.962; directed reachability traces 90.8 percent of value to cash-out. Legal mapping reports 100 percent only on high-confidence three-path agreement subsets of 8/8 and 15/15, with a 52.4 percent BM25 floor. These results are bounded measurements, not deployment evidence.

1. Requirements

- Keep complaint data and models under department control.
- Make every result reviewable and source-linked.
- Separate deterministic findings from generated language.
- Abstain rather than promote disagreement to a legal conclusion.
- Preserve a verifiable custody trail and explicit operator decisions.

2. Architecture

Paritran uses a deterministic graph and rules core behind local APIs. A separately fenced language component may draft cited text. Generated text cannot alter graph facts, custody records, or deterministic results.

Complaint-to-entity linkage

Identifiers extracted from synthetic complaint text resolve to shared entities. A graph joins complaints through common phones, devices, accounts, and other infrastructure. Communities become review candidates, not automatic allegations.

Money trail

A directed ledger represents movement from victims through mule layers toward cash-out. The 90.8 percent value-traced result is reachability over this synthetic ledger.

Human control

Operators see source records, confidence posture, and the reason for abstention. Acceptance and rejection are explicit workflow actions.

3. Legal mapping and grounded drafting

Rules, BM25 retrieval, and InLegalBERT each propose legal sections. Only agreement enters the high-confidence subset. Disagreement routes to an officer. This is precision via abstention, not a claim of universal correctness.

Measured mapping boundary

High-confidence agreement measured 100 percent on 8/8 and 15/15 cases. The BM25 v1 floor is 52.4 percent. Frozen benchmark values remain tied to their recorded environment; a dependency change requires re-measurement before equivalence is claimed.

F9 groundedness gate

The deterministic stub run recorded 40 claims passed, 10 withheld, and zero leaked. The live local gemma run recorded five passed, one withheld, and zero leaked. The gate checks support, but it does not replace legal review.

4. Experimental protocol and results

The demonstration uses synthetic data with planted ground truth and zero real PII. This permits direct measurement of linkage against a known answer key while avoiding a false claim of real-institution validation.

Measure	Result	Boundary
Linkage P / R / F1	0.957 / 0.966 / 0.962	Synthetic ground truth
Money trail	90.8 percent	Directed reachability
Legal mapping	100 percent, 8/8 and 15/15	Only when three paths agree
BM25 floor	52.4 percent	Condensed v1 corpus
F9 stub	40 / 10 / 0	Passed / withheld / leaked
F9 live gemma	5 / 1 / 0	Passed / withheld / leaked
NER P / R	0.74 / 1.0	Rule-augmented identifiers
Scale	297 complaints, 6/6 networks	Synthetic data
Custody	Verified and tamper-evident	Not immutable against full rewrite

5. Custody and security

- SHA-256 digests bind artifacts and records.
- An append-only hash chain makes local edits detectable.
- The chain head must be anchored outside the database to strengthen rewrite detection.
- Role boundaries and explicit sign-off keep the officer accountable.
- Local deployment reduces external dependency, but the operator must verify the actual network posture.
- Dependency audit runs without advisory suppression in CI.
- Synthetic-only publication prevents exposure of real complainant data.

Security non-claims

The prototype does not claim an HSM-backed production signing ceremony, independently certified isolation, real-data penetration testing, or immunity from a fully privileged host attacker. Those controls belong to a scoped pilot acceptance plan.

6. Limitations, deployment, and conclusion

Limitations

- Synthetic results do not establish performance on a police or bank dataset.
- The high-confidence legal result applies only to the stated agreement subsets.
- Tamper-evident does not mean immutable against full privileged rewrite.
- Drafting assistance does not establish admissibility or legal correctness.
- Production key custody, retention, deletion, and external anchoring remain TODO-VERIFY.

Pilot path

A pilot should use government-controlled hardware, approved source data, named operators, deterministic acceptance tests, rollback, physical verification, and signed custody receipts. Real data must remain outside the prototype until authority and minimisation controls are approved.

Conclusion

Paritran's contribution is not another opaque score. It is a narrow, reproducible path from linked synthetic complaints to a reviewable money trail, cited legal suggestions, a grounded drafting gate, and verifiable custody, with abstention and human authority preserved.

Canonical claim source: KANAD_SHIELD_2026_02_SOLUTION_DOSSIER.md
SHA-256: 800091205ad1e7c2a22a7add37f8d8fcfa1e5094f8737806bea56f5229fbc357